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Moral distress and ethical decision-making of eldercare professionals involved in digital service transformation

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ABSTRACT

Aim: Technology affects almost all aspects of modern eldercare. Ensuring ethical decision-making is essential as eldercare becomes more digital; each decision affects a patient's life, self-esteem, health and wellness.

Methods: We conducted a survey and interviews with eldercare professionals to better understand the behavioural ethics and decision making involved in the digital transition of eldercare.

Conclusion: Our qualitative analysis showed three recurrent roles among eldercare professionals in regard to digital service transformation; makers, implementers and maintainers. All three encountered challenging and stressful ethical dilemmas due to uncertainty and a lack of control. The matter of power relations, the attempts to standardize digital solutions and the conflict between cost efficiency and if digital care solutions add value for patients, all caused moral dilemmas for eldercare professionals. The findings suggest a need for organizational infrastructure that promotes ethical conduct and behaviour, ethics training and access to related resources.

IMPLICATIONS FOR REHABILITATION

- The transition to digital care service is not neutral, but value-laden. Digital transformation affects ethical behaviour and decision-making.
- The decision as to which digital services should be developed and deployed must include eldercare professionals and not lay solely in the hands of managers, technologists and economists.
- We must move away from attempting to fit standardized solutions to a heterogenous group of older patients; accommodating the pluralism of patients' needs and wants protects their dignity, autonomy and independence.
- As digital care practices evolve, so too must organizational structures that promote ethical conduct.

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Ethics; ethical decision-making; digital care services; eldercare professionals; moral distress

Introduction

Understanding ethical behaviour among eldercare professionals is vital, especially given the complexity of contemporary eldercare and its digital transformation. Caregiving involves constant ethics-based decisions [1] and addressing them may involve using theoretical and logic-based arguments to support an action [2]. However, everyday ethical behaviour among eldercare professionals is often intuitive and dependent upon the individual's beliefs and opinions of right and wrong in the given situation [3]. The concepts of ethics and morality are often used interchangeably [4], however, they have different meanings; ethics is the analysis, interpretation and systematization of questions of right and wrong while morality refers to specific beliefs, behaviours and ways of doing things [5]. The morals of an individual or group are manifested in their actions. An individual's morality is grounded in their culture and what is socially, legally and religiously acceptable therein [6].

In a profession like eldercare, a code of conduct drawn from agreed upon norms and values guides the actions of professionals within the field [7]. Christofferson (2007) argues that the code of conduct for care professions has three characteristics: (1) face-to-face meetings between people, (2) meetings that require some kind of action and (3) objective meetings that include actions based on the competence of the healthcare professionals involved [8]. However, increased use of technology in eldercare has changed these characteristics. Nowadays, meetings between patients and care professionals do take place not only at a single on-site location but also through conference calls, emails, digital devices and electronic platforms. Care professionals need to act on collected data regarding patients without actually interacting with them.

There are many devices that are part of the digital transformation of Swedish eldercare [9] including robots, sensors, GPS alarms and digital reminders [10–14]. These technologies provide digital surveillance, reminders, remote communication with care personnel as needed and physical examinations *via* cameras in the patients' homes [15]. Many digital eldercare services aim to enable older people to age in their own homes for longer than they could without digital services [16]. This development, in turn, leads to increased access to and better distribution of care, as

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well as to the re-allocation of responsibility [17–19]. Digital eldercare services alter the industry and foster new configurations of engagements and responsibilities [19]. Digital services in eldercare are rapidly becoming more common, which shows that technology affects human practices and experiences [20–22], and, therefore, ethical decision-making [23]. The digital transition in eldercare is already taking place and, as research publications on the ethics of digital care are still scarce, this article investigates how technology affects the ethical decision making of eldercare professionals and what, if anything, makes that decision making difficult [24].

The article is divided as follows: first, theories of ethics in digital care services contexts will be explained. They are used as informative concepts in the empirical aspects of this study. Then, the methodological approach taken will be described. After that, the results will be presented and discussed in the context of ethics and the digital transformation of eldercare by drawing on explanatory concepts from the literature. In the discussion, the results will be examined in the context of future digital care practices.

Thinking about ethics and digital eldercare services

Ethics can be defined as an active process of scrutinising and differentiating matters of right and wrong or good and bad [25]. Therefore, ethics, when referring to digital eldercare services, can be described as the process of deciding what kinds of digital services should be developed, implemented and used in eldercare, and asking questions as to whom and in what situations they should be provided, who benefits, who is marginalized or disadvantaged by their development, and what makes good care [11]. The process of making ethical decisions may be guided by formal ethical theories, ethical approaches or codes of conduct and is often called ethical decision making [25]. Regarding digital care services in the context of eldercare, the codes of conduct of the professionals involved are related to both the ethics of care and the ethics of technology. According to Noddings [26], the ethics of care involve both "caring-for" and "caring-about". Caring-for refers to the everyday practice of caregiving, within which there is a close relationship between the one who is cared for and the one who is giving the care. Caring-about refers to general ideas of right and wrong and what actions are considered to be good or bad at a societal level in regard to care. The difference lies both in the closeness of the relationship between the one who is cared for and the one giving the care and the level of implied responsibility. As Nodding states:

> The key, central to care theory, is this: caring-about (or, perhaps, a sense of justice) must be seen as instrumental in establishing the conditions under which caring-for can flourish. Although the preferred form of caring is caring-for, caring-about can help in establishing, maintaining, and enhancing it. Those who care about others in the justice sense must keep in mind that the objective is to ensure that caring actually occurs. Caring-about is empty if it does not culminate in caring relations. [26, p. 23–24].

The basic assumption (drawing on Beauchamp and Childress' model of ethics in care) is that both caring-for and caring-about ought to be guided by four major principles: (1) the principle of respect for autonomy, (2) the principle of non-maleficence, (3) the principle of beneficence, and (4) the principle of justice [27].

The principle of respect for autonomy refers to the idea of the individual's freedom from the control and interference of others; a person's basic right to make their own choices [28]. Autonomy as a right indicates that the individual's independence, participation and integrity ought to be respected by others [29]. According to the principle of autonomy in the context of eldercare in the digital age, the individual should have the right to choose not to use specific digital care services. Essentially, they should have the right to know what a particular service entails, how it functions and what consequences could arise from using it or refraining to use it, and then make their choices accordingly.

The principle of nonmaleficence refers to the idea that one should do no harm to patients [28]. This could mean, for example, refraining from unmotivated risk taking to minimize injuries and prevent suffering. In digital care practices, the principle of nonmaleficence may result in withdrawing a digital service if the disadvantages, such as distress, anxiety or social isolation are disproportional to the benefits, such as efficacy and effectiveness, or refusing to apply a digital service that has no proof of success or efficacy.

The principle of beneficence refers to the specific aim to do good. Eldercare professionals should strive to do good by meeting the needs of patients, both medical and social, in order to ensure the patient's wellness and quality of life [28]. In terms of digital care, this principle suggests that caregivers should evaluate the impact of a particular technology on the patient's health and wellness. Replacing physical homecare visits with digital homecare visits, for example, may or may not meet the physical and social needs of the individual patient. The patient's best interests, not the economic goals of the care organization or its desire to seem forward-thinking, should guide choices regarding digital solutions and their application. Mahoney et al. mention that beneficence in regards to digital services also applies to personnel who introduce and install the technologies in a patient's home and that they should be trained to interact with patients with different kind of needs [30] in order not to upset them or cause anxiety.

The principle of justice refers to the idea that each patient should be treated fairly. Daniels [31] argues

> Key elements of fair process will involve transparency about the grounds for decisions; appeals to rationales that all can accept as relevant to meeting health needs fairly; and procedures for revising decisions in light of challenges to them. Together these elements assure "accountability for reasonableness.
>
> [31, p. 1316]

Essentially, the care provided should be fair and free from bias and prejudice [28]. Digital services should be distributed on just and equal grounds and in proportion to the extent of the patient's need of care. No patient group should be disadvantaged or particularly benefitted by digital care [32]. Patients with similar needs should be offered the same kinds of digital care services independent of their socio-economic status or whether they live in rural or urban municipalities. Mahoney et al. highlight the importance of developing digital care services that are affordable for all who may benefit from them [30].

Beauchamp and Childress' principles are *prima facie*, which means that they ought to be followed and respected but also allow exceptions when they conflict with each other [27].

Several models have been suggested to ethically evaluate technology [33,34]. Since different technologies can be in different phases of development, at any point from the idea phase to that of the end product, different ethical evaluation models address different ethical aspects. Brey proposes the anticipatory technology ethics (ATE) model, which consists of three phases: (1) foresight analysis, (2) the identification of ethical issues and (3) the evaluation of ethical issues [35]. According to Brey, ATE can be used to make projections for possible ethical issues regarding emerging technologies and their applications [36, p. 138]. These

insights can be fed back into the design process and can thereby affect development. The goal is anticipating future ethical consequences as the technology is used. It is not easy to anticipate all ethical outcomes and it is recommended that ethical analyses are carried out iteratively during development. The ATE approach offers a generative way to integrate ethics into the development of health and welfare technologies, as it can support both creativity and novel problem-solving. Brey also emphasizes the importance of initiating ethical analysis at the earliest possible stage of development in order to shorten the social backlog as much as possible and to ensure that the technology is designed in accordance with societal norms and values [35].

When a technology is already developed and ready to enter an eldercare context, an ethical risk analysis can be used to help determine if the technology should be implemented, by whom, in which situations, and who has the potential to benefit or be disadvantaged. Hansson [37] proposes a three-step ethical risk analysis method: (1) identify and categorize concerned people into the three roles of risk-exposed, beneficiaries and/or decision-makers; (2) analyse the roles and their potential combinations in regards to ethical risks and (3) deliberate on individual risk-benefit weighing, distributional analysis, rights analysis and power analysis. That is, one must identify if there is a moral problem by obtaining information about a specific health and digital care service and its proposed application, background and technical, medical and legal facts. The different affected parties and their respective interests and needs must also be identified and compared. The next step is to identify one or more technological option(s) and to assess the consequences of each one for each party in the short and long term. Furthermore, the consequences of doing nothing (maintaining the status quo) should be assessed, ending with ethical reasoning in relation to ethical theories such as utilitarianism, deontology and virtue theory. By using several different ethical theories, many different perspectives and possible solutions will emerge [34]. Ethical risk analysis is helpful in envisaging pluralism in terms of ethical consequences related to the affected actors' perspectives, needs, wants and contexts and in mapping it to different kinds of solutions/actions.

This brings us to the empirical question: What kind of ethical decision-making takes place on a day to day basis as eldercare becomes more digital?

Methodological approach

This study draws on the results of a survey and in-depth interviews with willing respondents to explore ethical behaviour in regard to municipal eldercare services transitioning to digital alternatives.

A survey was sent out in March 2020 to those responsible for digital service transformation in eldercare in 40 municipalities. The survey was developed using Google forms. The respondents were informed that their participation was anonymous and voluntary, that the data files with their answers were confidential, that the answers could not be linked to an individual and about the aim of the questionnaire (to explore ethical decision making in the everyday practice of digital care service transformation). The first question was: "Do you encounter any ethical matters in regard to digital care services in your everyday work practices"? If they answered yes, exploratory open-ended questions followed concerning what kinds of ethical matters were faced, specifically which kinds of digital services, patients and situations and their influence on ethical decision-making.

Qualitative data analyses were performed using NVivo software. The data were analysed according to the principles of qualitative data analysis with inductive and deductive phases [38]. The survey answers were read with a focus on the respondents' words and how they framed their views of their work in relation to ethics, caregiving and the digital transformation of eldercare. Significant statements were extracted and assembled [38].

The results of the survey were used to develop the questions for the ten conducted in-depth interviews. They were conducted according to an interview protocol and each interviewee was asked the following: (1) How would you describe your role in the digital service transformation of your eldercare organization; (2) please tell me about different situations involving digital care services when you were not sure about what the right thing to do was regarding patient autonomy, non-maleficence, beneficence, and justice; (3) how were the situations resolved; and (4) what kind of resources do you use to help you resolve ethical problems. Prompts and probes were used to make the interviewees expand on the answers (e.g., could you please give me an example on X, Y and Z; what do you mean when you say X, Y and Z; tell me a little bit more about X, Y and Z, etc.) [39]. The interview protocol acted as a guide to help the interviewer focus on ethical behaviour and digital care services. The interviews lasted approximately 1 h and were conducted in Swedish by the author. The interviews were recorded, transcribed and then analysed according to the principles of qualitative data analysis with inductive and deductive phases [38]. The process involved identifying and coding data related to ethical behaviour and digital services. The next step was synthesizing the data from the survey with the interviews with the intent of clustering them into themes. These data were then grouped into larger themes, incorporating all mentions of ethical behaviour in the everyday practices of digital care services and the related ethical challenges and dilemmas. The themes were reviewed by the interviewees and discussed until concordance was reached.

The guidelines of research ethics issued by the Swedish Research Council [40] were followed. The research does not shield any sensitive information that can be linked to an individual and, therefore, does not require ethical approval according to the Swedish regulations on research ethics [40].

Results

The participants

The online survey resulted in a total of 40 responses from 40 different municipalities. A hyperlink to access the online questionnaire was sent to the registrar in 40 municipalities. The registrar was asked to distribute the link to eldercare professionals responsible for municipal digital service transformation. Five respondents answered that they did not encounter ethical dilemmas regarding digital care services in their everyday work practices, and so did not complete the survey, while 35 respondents did. The titles of the respondents varied: head of unit (10%), digitalization strategist (30%), quality assurance developer (5%), chief nurse (20%), eHealth and digital care services coordinator (30%) and business developer strategist (5%). Though their titles were different, they were all responsible for digital service transformation in their eldercare organizations. Most of the respondents were female (85%), confirming the notion that healthcare is a female-dominated occupation [41,42]. Their ages ranged from 28 to 65 years old, with the majority aged from 42 to 54 years old. All but one had a university degree.

The in-depth interviews were conducted with ten people responsible for digital care service transformations in ten different municipalities. The interviewees were respondents to the survey who were willing to being interviewed and included eight women and two men. Their ages ranged from 30 to 58 years old, with the largest group consisting of those aged between 40 and 50 years old. The interview subjects worked within public municipal eldercare in both rural and urban settings.

The roles of eldercare professionals within the field of digital care service transformations and implications for ethical decision-making

The study reveals that people responsible for municipal digital care services can take on three roles in regard to digital service transformation: (1) makers (members of the development team), (2) implementers and/or (3) maintainers. The thematic analysis revealed that these three roles are recurrent and reproduced fair consistently across a range of municipalities. An individual can take on more than one of these roles depending on their work situation. Each of the three roles and its linked ethical behaviour involve actions and responses from several municipalities and individuals (Figure 1).

Eldercare professionals as makers of digital care services

Eldercare professionals sometimes have the opportunity to participate in the development of emerging digital care services through both nationally and internationally funded technological development projects, and can thereby be seen as makers; part of the development team [43,44]. In this context, the technologies are still in the making and eldercare professionals can affect the direction the development takes before the technological solution becomes embedded in routines that may be problematic to alter. When it comes to ethical issues and developing technologies, ATE can be used. This involves (1) foresight analysis, (2) the identification of ethical issues and (3) the evaluation of ethical issues [33,35]. Several of the participants mentioned that they had been a part of projects aimed at developing new digital care services. They noted that new digital services challenge actions and points of view that were previously taken for granted. They acknowledged that they regularly carried out technology forecasting, evaluating their current care methods in comparison with potential future digital ones. As one digitization strategist wrote:

> Emerging technology challenges accepted assumptions, norms, and values – we need to question if what we do today is more or less ethically correct in comparison with a digital solution.

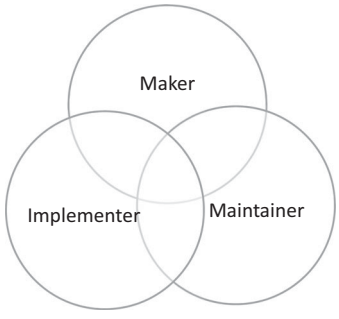


Figure 1. Different roles of eldercare professionals in the field of digital care services.

The ethics of makers of digital care services, as shown in the interviews and survey, were related to recognizing that digital care services might bring about positive change and comparing everyday practices to future technological opportunities. A recurring mantra among the participants was, 'good and safe care, to do good for the patient'. The actual impact that eldercare professionals have on the design process was questioned [45,46]. As one of the interviewees explained:

> We may work together with technology developers, doing pre-studies and involving our patients, but our actual impact on the design process I am not sure about? … in some ways we are in the hands of the technology suppliers … they have a core range of digital services that they offer and if we need adjustments to our needs or other kinds of services, we can sometimes get them if we pay extra, but it is more of a standardised universal package than a smorgasbord of options … maybe one could say that we are involved in development projects more as test sites than actual developers … some technology suppliers are more responsive to our input than others.

The lack of power expressed above was experienced by several of the participants who had been a part of development teams. Their ethical behaviour as makers was restricted by technology suppliers, contract agreements and IT departments in regard to what kind of technological solutions they could design and develop. This raises ethical questions about who participates in and has control of the development of digital care services.

Eldercare professionals as implementers of digital care services

Another role identified during the analysis of the survey and interviews was the implementer. As implementers of digital care services, eldercare professionals consider ethical issues in regards to the use of a particular digital care service in specific situations, practices or contexts [47,48]. In these circumstances, they may need to either trust their moral insights and intuition or conduct an ethical risk analysis [37]. Although none of the survey respondents or interviewees mentioned ethical risk analysis, many mentioned cost-benefit analyses. Cost-benefit analyses involve comparing the costs of different digital services in relation to the potential benefits to patients, personnel and the organization. As one of the interviewees explained:

> All the time we need to explore which option is most value-creating in relation to the costs of that option … we have a very tight budget and an increase in patients … I have to prioritise and decide what can be done within that budget … I do not always choose to implement the best solution but hopefully a solution that is good enough.

There was a notable ethical awareness among the survey respondents and interviewees regarding digital care service implementation. Several indicated that digital care service implementation should add value for both patients and eldercare professionals. However, in the interviews, it became clear that the role of implementer involved several moral challenges. In this case, a moral challenge is a choice between two or more undesirable consequences of technological solution implementation when they do not feel morally good or right.

The interviewees offered some examples as follows:

> I found this great system that I thought was going to be used to help homecare professionals to manage all their home visits to patients. My intention was that the system would ease the work burden for the personnel as well as ensuring that each patient got the services they were entitled to. The software enabled ways for easier and quicker planning and keeping track of and scheduling daily tasks but after implementation the system was used to increase efficiency and supervise the eldercare professionals' everyday activities – minute by minute … in the end, the digital system actually increased the stress

> levels of the eldercare professionals while the deviations of care efforts decreased ….

> In the past, we most often put diapers on elderly patients when they entered our elderly care institution … I thought that this was not right and found a digital solution … it was a chip that could monitor urination … my idea was that if we developed an understanding of each patient's individual urination pattern, we could use this knowledge to take them to the toilet at the right time instead of them wearing diapers … however, sometime after the implementation I found out that the intervention was used to decide what kind of diaper … the thickness … was needed for each patient instead of actually removing diapers and encouraging patients to use the bathroom … instead, the solution was used to find the diaper that lasted the longest and did not need to be changed as often …

> I decided that instead of having all the paper lists with what kind of tasks they [homecare personnel] should do in each patient's home, they could get the information sent digitally to their mobile phones … this would help them to keep track of the addresses of the patients and their daily activities … but after the implementation a lot of the patients complained that the caretakers were paying more attention to their phones than to them … the older patients felt neglected and invisible …

> In our municipality, I was responsible for doing a pilot of digital night surveillance … about seven patients volunteered to try it out … during the evaluation, it turned out that all the participants were very happy with digital surveillance … their sleep was not disturbed like when they had physical night visits … after the pilot finished, all the participants wanted to keep the digital surveillance – a good outcome. We decided to procure digital surveillance and I thought patients would have their say as to whether they wanted digital surveillance or physical night visits, but the local politicians decided the only choice offered from now on would be digital visits … first, the digital care service was viewed as one solution of many to a problem or need, and then it became the only solution …

> In my role, I have the responsibility of finding technological solutions that enable older people to live in their own homes for as long as possible … but I often ask myself if it is what they want. Or do we just try to use technology to make them stay at home for as long as possible because moving them into special institutions is more expensive? … we are trying to find solutions to suit all older people but in reality it should be dealt with case by case … some feel safer at home, and then technology may be used as security, but others are worried, lonely, and do not feel safeguarded by having different kinds of technological solutions …

These interview excerpts are only a few of the many examples of the challenges and ethical dilemmas that the implementers of digital care services have encountered. Many expressed frustrations and indications of moral distress due to their inability to control the effects of the digital services and how they were actually implemented. As is often the case with technology, there is great optimism about what it can do, but, as learnt from science and technology studies, technology both shapes and is shaped by the context in which it is implemented [49,50]. The ethical and social effects of technology are hard to anticipate, however, a systematic and in-depth analysis would most likely help reveal several of the ethical consequences mentioned above. None of the survey or interview participants mentioned that they used any formal ethical assessment methods in the implementation phase. When prompted, some participants mentioned that they evaluated ethical issues when they arose in multi-professional teams or in dialogue with the patient and/or their relatives and other affected care professionals. Others searched for information within digital care service networks and the experiences of other municipalities. Still, others looked to laws and regulations to guide their decisions and ethical considerations, reasoning that what is legal must also be ethically correct. By following laws and regulations (e.g., the principle of justice), they felt safeguarded in their ethical behaviour. However, some of the respondents to the survey, as well as the interviewees, highlighted the Social Services Act as an example of a framework law that lays down general obligations and principles but leaves decisions regarding how digital care services should be implemented up to municipalities. Therefore, many eldercare professionals felt more confident introducing digital care solutions that had already been implemented in other municipalities. Other respondents stated that laws and regulations felt outdated in regard to the digital transformation of eldercare and had negative effects on moral decisions because they offered no guidance. For example, one respondent wrote:

> When introducing the idea of implementing robots that handle medical drugs in the homes of patients, the question arises as to if the robots are considered as homecare or self-care. In this case, the legislation was based on old principles and not very applicable … homecare is paid for by the municipality while self-care is paid for by the patient himself/herself … since we could not decide if the robots were home-care or self-care, we ended up doing nothing … in such circumstances, one could say that the laws and regulations hindered development …

Eldercare professionals as maintainers of digital care services

As the use of technology in eldercare is constantly increasing [11,19,51–56], the most common role eldercare professionals play is neither maker nor implementer, but maintainer of digital care services [52,57–60]. In this context, technology is already embedded in everyday routines and work practices. Consider mobile safety alarms with GPs as an example. Their implementation changes the responsibility of eldercare personnel from being in charge of the patient's wellness at institutions or in their homes to being responsible for making sure that patients wear the mobile safety alarm all the time and for reacting to the output of the system. They also need to check that the batteries in the devices are always charged and that the technology is functioning.

In the everyday practices of eldercare professionals as maintainers of digital care services, their ethical behaviour is often implicit. Initially, they did not acknowledge these factors as ethical behaviour but rather as the right thing to do or a work process that they followed. For example, when asked about ethical matters, some of the respondents answered that they did not encounter any in regards to maintaining digital care services. However, when prompted to elaborate, they started to share experiences of their everyday work practices, and it became apparent that much of their ethics related behaviour occurred on a daily basis. Their ethical behaviour was closely related to Beauchamp and Childress' four principles – autonomy, nonmaleficence, beneficence and justice. In most municipalities an initial needs assessment is carried out before a patient can get a certain digital care service solution. The initial needs assessment is performed using a framework called the Individual Needs Based Assessment in which the individual's needs are the focus. However, matching the individual's needs with a digital care service solution is often an ethical challenge. As one of the participants said:

> We do this quite extensive individual-based assessment to find out all the needs of a patient … both social and medical needs … but then we have to shrink these needs to fit a standardised digital care service solution … we do not have a variety of digital care services to choose from and therefore it feels like we sometimes have to convince both ourselves and the patient that his/her needs are met by the digital care services device … while people such as homecare personnel can adjust their behaviours towards the patient's needs, digital care service solutions are not very adjustable but are fixed in terms of the ways in which they can be used and interacted with …

If it is decided that the patient needs digital care services, they then must consent to their use. Questions were raised among our participants about ethical issues concerning people with cognitive impairments and digital care services. Often, patients with severe cognitive impairments do not understand what the technology does and what they are consenting to. Informed consent has been the most cited ethical concern when it comes to technology development and people with cognitive decline [11,61,62]. Depending on the patient's diagnosis, family members can become surrogate decision makers [63]. In many cases, the interviewees felt that they had to make the decisions together with relatives regarding digital care services for people with severe cognitive decline but were unsure who benefitted the most from the technology. They said that it was the relatives or the home-care professionals that suggested the technology and that the patient with cognitive decline was left out of the decision making. They acknowledged that surveillance technology might result in unethical infringement on a patient's bodily integrity. However, they also argued that surveillance technology was primarily used to ensure the good health and security of the patient (e.g., non-maleficence (do no harm) and beneficence (do good)). Niemeijer et al. argue that eldercare professionals need to base their ethical decisions on the balance between the risk of harm to patients and the right to autonomy [64].

It became clear that ethical questions were more in the background than in the foreground for maintainers, since the digital care services had already been procured and implemented. As one of the interviewees explained:

> There is a risk that, having gone through several digital care services implementations … you lose sight of ethical issues … today, digital care services are accepted without question or without any critical thinking … you just do it … thus it is maintained … that's what society looks like … we use technology for everything.

Moral distress

This exploration of ethical behaviour exposed moral distress associated with municipal digital care service transformation. Moral distress refers to challenging ethical situations that give rise to distress [65]. There was an overlap between the causes of moral distress and the different roles found in the analysis (makers, implementers and/or maintainers). Two main factors related to moral distress in digital care service transformation were identified: (1) uncertainty and (2) lack of control.

Uncertainty

In order to react to an ethical situation, one has to be aware that ethical concerns may arise in the development, implementation or maintenance of digital care services and must acknowledge that ethical problems have potentially already surfaced in practice [66,67] define ethical awareness as competence in recognizing ethical implications in everyday healthcare practices and as the first step towards moral action [67]. In the context of this study, the participants acknowledge ethical principles of care including autonomy, nonmaleficence, beneficence and justice. They highlighted the importance of assessing the patient's individual needs in order to offer flexible care services to meet them. They felt reassured because municipal eldercare organizations already had work routines and formalized procedures for individual-based assessments. However, in the context of digital care services, knowing the appropriate assessment procedure seems to be more challenging. The participants felt unprepared to manage the ethical challenges that digital care services impose on care practices. Reasons for this appear to be grounded in a lack of knowledge and awareness about different digital care options and the consequences of using different kinds of solutions as well as a lack of understanding as to which technological solution is both best for a patient's needs and legally available to the provider. Often, a way of getting around the implications of not having the right tools for a truly ethical assessment of digital care services in relation to individual patients was to implement digital care services that other municipalities had already used. During such instances, the moral responsibility of the technology was shifted from the organization to other municipal eldercare organizations and their previous decisions. Even though the participants followed the strategy of implementing technology that was already implemented in other municipalities, they raised concerns about informed consent and people with cognitive decline. They questioned how to best understand the patient's wishes and to make sure that the patient understands what they are consenting to. As one of the participants expressed:

> How do I know that digital night surveillance is the right option for my patient? I know that in other municipalities they use cameras for people with cognitive decline, but my patient does not understand that she is being digitally supervised. The digital surveillance might keep her safe, but I can't tell what the best option is … but in the end, I still persuaded her to use digital night surveillance because it seems to work for others with similar problems to hers …

The participant quoted above expressed concern regarding digital surveillance and her patient. As was the case with many of the participants, she was aware that the technology raised ethical and legal issues but was not aware of the 'correct' course of action and decided to do 'what everyone else would do', in this case prescribe digital surveillance. In the municipality in which she was employed, you either prescribed digital surveillance or there was no night surveillance at all. There was no other option. According to the Beauchamp and Childress principles, she should respect the autonomy of the patient and at the same time pursue good and avoid doing harm [28]. In these circumstances (described in the extract), she had to renounce her ethical values because she lacked the knowledge and tools to deal with the situation in a way that was satisfactory to her, which, in turn, caused uncertainty and distress as to whether she made the right decision.

Lack of control

A structural condition that affected the everyday experiences of eldercare professionals implementing digital care services was lack of control over the end result (e.g., usage and application), which the interviewees attributed to various factors, such as responsibility, power and societal changes. The participants felt that even when they were involved as makers their participation did not result in any major input on the final solution. Technology suppliers, contract agreements and the municipal IT departments made the decisions regarding viability and development potential. The responsibilities of the eldercare personnel involved in developing digital care services were to provide information and examples of everyday care situations and run test sites. Even though ethical issues were raised, they felt that they did not have enough developmental power to give them priority.

In the implementation phase, the interviewees felt that even when they initially thought that they had made the right decision and had chosen the right technology to implement, they later questioned their actions since the technology was diffused in a

way that was different from what they had had in mind. In these circumstances, they felt that they had a moral responsibility to implement the right technology but that they lacked control over how the technology was interpreted and used in everyday care practices.

Another structural factor which affected the participants' moral responsibility was the digital transformation of society. Seeing technology as a driving force for organizational change caused moral distress since the participants worried about whether the changes were for the benefit of the patients or if the main organizational motive was financial. At the same time, the moral responsibility for the digital transformation of municipal eldercare shifted from eldercare organizations to society. For example, one of the participants expressed the following: 'Society is changing, everything is getting digitalised and so is municipal eldercare'. Ethical risks were neglected in order to succumb to the pressure to transform municipal eldercare into a digital service.

Discussion and implications

This study examined ethical behaviour among Swedish eldercare professionals who were responsible for digital care service transformation in municipal eldercare. Eldercare professionals can take on three roles in relation to ethical behaviour and digital care services: makers, implementers and maintainers. In relation to these roles, this study identifies three clear trends affecting ethical behaviour among eldercare professionals involved in digital service transitions:

Power relations

Makers were often ethically constrained by their lack of power in relation to technology suppliers, contract agreements, procurements and the municipal IT departments. The issue of who has the power in digital transformation decision making has been in explored in other empirical research [68,69]. Their research shows that although digital transformation of care may enable changes in power structures, there seems to be evidence that existing power structures remain or even are reinforced, as it is the engineers, technologists and economists that make the final decisions, not the elder care professionals [68,69].

The eldercare professionals were able to better follow their moral instincts and moral obligation to their patients as implementers, however, they lacked control over how the technological solutions were actually diffused. The practice of maintaining digital care services in everyday care was filled with uncertainty as to the right thing to do and what exactly a good life with digital services for eldercare patients is.

Standardized digital solutions versus *individual needs*

The results imply that digital care service transformation gives rise to challenging ethical situations that eldercare professionals feel unprepared to handle. One such challenge was fitting the uniformity of digital care services to the pluralism of patients' needs and wants. The fact that municipalities procured digital devices and digital service platforms in bulk for what seemed to be a homogenous group of patients conflicted with the goal of providing care services based on the patient's individual needs. The eldercare professionals felt compelled to coerce patients into accepting digital services and devices because that was all they had to offer, as opposed to starting from the results of the individual-based assessment and finding a desirable solution with the patient. The implications of reducing patients' capabilities and trying to match their impairments to a standardized digital service caused moral distress. The implications of reducing disability to a single dimension have been significantly highlighted in the field of disability studies [70–72]. Still, the eldercare professionals in this study regularly needed to address this matter. This matter of concern and call to consider technology as not just a device or digital service to "fix" an impairment but rather as something that should fit into and support the quality of life of people in need of care has been highlighted by several researchers [73–77].

Cost efficiency versus added value

Technology is not neutral [78,79], as it is developed by humans with certain aims and purposes in mind. The design of a technology or digital service affords certain actions while it restrains others [79,80]. At the same time, different people can interpret and give different meanings to the same technology [50,81]. Many digital care services for older people are developed to monitor and manage care [30]. According to Robinson et al., the principles of non-maleficence (do no harm) and safety is in the foreground when it comes with people with dementia [82]. Other researchers indicate that the digitizing of eldercare services is pushed and promoted due to political motivations regarding cost efficiency in healthcare [83–85]. Interestingly, there is no clear evidence that the digital care transition is cost effective [86]. The eldercare professionals in this study unfortunately seemed torn between organizational interest in becoming digital and their duty of care for the patients. For them, their obligations towards the patient and their employers were perceived as conflicting and morally problematic. Several were unsure if the digital services really added value to the lives of the patients and protected their dignity, autonomy and independence or if they were introduced for the benefit of the organization. Beauchamp and Childress' principles (autonomy, non-maleficence, beneficence and justice) point towards the idea that no digital solutions will meet the needs of all patients. In order to follow and respect the principles, digital solutions need to prioritize the patient's wishes, wants, needs, health and wellness.

Conclusion

There is not one factor alone that can be used to understand how and why digital care services cause challenging ethical situations that give rise to distress. The reasons clearly extend beyond the individual to organizational and economic rationality, policy and societal influences. However, one thing is clear; there is a need to raise ethical awareness and develop ethical competence in the everyday practices of digital care service transformation. This might be done through guidance, standards, procedures and infrastructures that enable ethical behaviour, training and access to ethics and digital care services related resources. As digital care practices evolve, so too must organizational structures that promote ethical conduct.

Limitations of the study and future research

One limitation with this study is that it does not include the perspective of the patients, their family members, technologists and policymakers. One could explore, for example, how digital care services are introduced to older patients and their family members. One could investigate if and how digital services protect the older patient's dignity, autonomy and independence, if at all. Are patients able to use digital care services? Why are they motivated or, by contrast, reluctant to use a digital service? and so forth. Interviews or surveys with technologists would focus on design, seeking an insight into decision making and which experiences of older patients influence these design decisions. Interviews or surveys with policymakers would help researchers understand the processes, tensions and trade-offs that are currently seen in municipality procurement, distribution and employment of digital care services. All these different perspectives are, of course, important for pluralism in terms of ethical matters related to the digital transformation of eldercare.

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References

[1] Tuckett AG. Virtuous principles as an ethic for nursing. Contemp Nurs. 2000;9(2):106–114.
[2] Cronqvist A, Theorell T, Burns T, et al. Caring about-caring for: moral obligations and work responsibilities in intensive care nursing. Nurs Ethics. 2004;11(1):63–76.
[3] Puig de la Bellacasa M. Ethical doings in naturecultures. Ethics Place Environ. 2010;13(2):151–169.
[4] Sandman L, Kjellström S. Etikboken: etik för vårdande yrken [The book on ethics: ethics for care professionals]. 2013.
[5] Beauchamp T. Philosophical ethics: an introduction to moral philosophy. 2001.
[6] Kohlberg L. Moral stages and moralization'. Moral Dev Behav. 1976;5:31–53.
[7] Somers MJ. Ethical codes of conduct and organizational context: a study of the relationship between codes of conduct, employee behavior and organizational values. J Business Ethic. 2001;30(2):185–195.
[8] Christoffersen SA. *Professionsetik* (Gleerups). 2007.
[9] Löfqvist C, Slaug B, Ekström H, et al. Use, non-use and perceived unmet needs of assistive technology among Swedish people in the third age. Disabil Rehabil Assist Technol. 2016;11(3):195–201.
[10] Lewin D, Adshead S, Glennon B, et al. Assisted living technologies for older and disabled people in 2030', A final report to Ofcom. London: Plum Consulting; 2010.
[11] Hofmann B. Ethical challenges with welfare technology: a review of the literature. Sci Eng Ethics. 2013;19(2):389–406.
[12] Sánchez VG, Taylor I, Bing-Jonsson PC. Ethics of smart house welfare technology for older adults: a systematic literature review. Int J Technol Assess Health Care. 2017; 33(6):691–699.
[13] Coeckelbergh M. Artificial agents, good care, and modernity. Theor Med Bioeth. 2015;36(4):265–277.
[14] Stokke R. "Maybe we should talk about it anyway": a qualitative study of understanding expectations and use of an established technology innovation in caring practices. BMC Health Serv Res. 2017;17(1):657.
[15] Axelsson K, Harrefors C, Sävenstedt S, et al. E-Health in care of older persons in the future. J Clin Nurs. 2010;19:54.
[16] Kolkowska E, Avatare Nöu A, Sjölinder M, et al. To capture the diverse needs of welfare technology stakeholders–evaluation of a value matrix. In International conference on human aspects of IT for the aged population. Berlin, Germany: Springer; 2017. p. 404–419.
[17] Clarke AE, Shim JK, Mamo L, et al. Biomedicalization: technoscientific transformations of health, illness, and US biomedicine'. Biomed Technosci Health Illn. 2003;47–87.
[18] Nam S, Han SH, Gilligan M. Internet use and preventive health behaviors among couples in later life: evidence from the health and retirement study. Gerontologist. 2019;59(1): 69–77.
[19] Frennert S, Östlund B. Narrative review: technologies in eldercare. J Adhesion Sci Technol. 2018;6(1):21–34.
[20] Verbeek PP. Materializing morality: design ethics and technological mediation'. Sci Technol Hum Values. 2006;31: 361–380.
[21] Verbeek PP. Beyond interaction: a short introduction to mediation theory. Interactions. 2015;22(3):26–31.
[22] Verbeek PP. Toward a theory of technological mediation. Technoscience and Postphenomenology: the Manhattan. Lanham (MD): Lexington Books; 2015. p. 189–204.
[23] de La Bellacasa MP. Matters of care in technoscience: assembling neglected things. Soc Stud Sci. 2011;41(1): 85–106.
[24] Brall C, Schröder-Bäck P, Maeckelberghe E. Ethical aspects of digital health from a justice point of view. Eur J Public Health. 2019;29(3):18–22.
[25] Sidgwick H. The methods of ethics. Glasgow: Good Press; 2019.
[26] Noddings N. Starting at home: caring and social policy. Berkeley (CA): University of California. Press; 2002.
[27] Beauchamp T, Childress J. Principles of biomedical ethics. New York (NY): Oxford University Press; 2001.
[28] Childress JF, Beauchamp TL. 2001. Principles of biomedical ethics. New York (NY): Oxford University Press.
[29] Linzer N. An ethical dilemma in home care. J Gerontol Soc Work. 2002;37(2):23–34.
[30] Mahoney DF. An evidence-based adoption of technology model for remote monitoring of elders' daily activities. Ageing Int. 2011;36(1):66–81.
[31] Daniels N. Accountability for reasonableness: establishing a fair process for priority setting is easier than agreeing on principles. BMJ. 2000;321:1300–1301.
[32] Meiland F, Innes A, Mountain G, et al. Technologies to support community-dwelling persons with dementia: a position paper on issues regarding development, usability, effectiveness and cost-effectiveness, deployment, and ethics. JMIR Rehabil Assist Technol. 2017;4(1):e1.
[33] Brey P. Anticipating ethical issues in emerging IT. Ethics Inf Technol. 2012;14(4):305–317.
[34] Hofmann B. Hvordan vurdere etiske aspekter ved moderne helse-og velferdsteknologi? Tidsskr Omsorgsforsk. 2019; 5(03):99–116.
[35] Brey P. Anticipatory ethics for emerging technologies. Nanoethics. 2012;6(1):1–13.
[36] Brey P. Philosophy of technology after the empirical turn. Tech Res Philos Technol. 2010;14:36–48.
[37] Hansson SO. How to perform an ethical risk analysis (eRA). Risk Anal. 2018;38(9):1820–1829.
[38] Graneheim UH, Lundman B. Qualitative content analysis in nursing research: concepts, procedures and measures to achieve trustworthiness. Nurse Educ Today. 2004;24(2): 105–112.
[39] Kwortnik RJ, Jr. Clarifying "fuzzy" hospitality-management problems with depth interviews and qualitative analysis. Cornell Hotel Restaurant Administrat Quart. 2003;44(2): 117–129.
[40] Council, Swedish Research. Good research practice. Stockholm, Sweden: Swedish Research Council Stockholm; 2017.
[41] Hartmann H, Hayes J. The growing need for home care workers: improving a low-paid, female-dominated occupation and the conditions of its immigrant workers. Public Policy Aging Rep. 2017;27:88–95.

[42] Hayes LJB, Moore S. Care in a time of austerity: the electronic monitoring of homecare workers' time'. Gender Work Organ. 2017;24:329–344.
[43] Kushniruk A, Nøhr C. Participatory design, user involvement and health IT evaluation. Stud Health Technol Inform. 2016; 222:139–151.
[44] Grönvall E, Kyng M. On participatory design of home-based healthcare. Cogn Technol Work. 2013;15:389–401.
[45] Woelders S, Abma T. Participatory action research to enhance the collective involvement of residents in elderly care: about power, dialogue and understanding. Action Res. 2019;17(4):528–548.
[46] Grosjean S, Bonneville L, Redpath C. The design process of an mHealth technology: the communicative constitution of patient engagement through a participatory design workshop. ESSACHESS J Commun Stud. 2019;12:5–26.
[47] Ericson Lidman E, Strandberg G. Care providers' troubled conscience related to an implementation of a time management system in residential care for older people—a participatory action research study. Scand J Caring Sci. 2020;34(3):745–753.
[48] Pekkarinen S, Melkas H. Welfare state transition in the making: focus on the niche-regime interaction in Finnish elderly care services. Technol Forecast Soc Change. 2019;145:240–253.
[49] MacKenzie D, Wajcman J. The social shaping of technology. London: Open University Press; 1999.
[50] Bijker WE, Huges TP, Trevor P. Social construction of technological systems. Cambridge: MIT Press; 2012.
[51] Kamp A, Obstfelder A, Andersson K. Welfare technologies in care work. Nordic J Work Life Stud. 2019;9(S5)12.
[52] Ertner M. Enchanting, evoking, and affecting: the invisible work of technology implementation in homecare. Nordic J Working Life Stud. 2019;9:33–47.
[53] Cozza M. Interoperability and convergence for welfare technology. International conference on human aspects of IT for the aged population. Las vegas (NV): Springer; 2018. p. 13–24.
[54] Zander V, Johansson-Pajala RM, Gustafsson C. Methods to evaluate perspectives of safety, independence, activity, and participation in older persons using welfare technology. A systematic review. Disabil Rehabil Assist Technol. 2019;15: 373–393.
[55] Hansen AM, Grosen SL. Transforming bodywork in eldercare with wash-and-dry toilets. Nordic J Work Life Stud. 2019;9:49–67.
[56] la Cour A, Højlund H. Untimely welfare technologies. Nordic J Work Life Stud. 2019;9(S5):69–87.
[57] Sánchez-Criado T, López D, Roberts C, et al. Installing telecare, installing users: felicity conditions for the instauration of usership. Sci Technol Hum Values. 2014;39:694–719.
[58] Mol A, Moser I, Pols J. Care: putting practice into theory'. Care in practice: on tinkering in clinics, homes and farms. Vol. 8. Bielefeld, Germany: Transcript Verlag; 2010. p. 7–27.
[59] Pols J. Wonderful webcams: about active gazes and invisible technologies'. Sci Technol Hum Values. 2011;36: 451–473.
[60] Pols J, Willems D. Innovation and evaluation: taming and unleashing telecare technology'. Sociol Health Illn. 2011; 33(3):484–498.
[61] Marziali E, Serafini JMD, McCleary L. A systematic review of practice standards and research ethics in technology-based home health care intervention programs for older adults. J Aging Health. 2005;17(6):679–696.
[62] Robillard JM, Cleland I, Hoey J, et al. Ethical adoption: a new imperative in the development of technology for dementia. Alzheimers Dement. 2018;14(9):1104–1113.
[63] Fetherstonhaugh D, McAuliffe L, Bauer M, et al. Decision-making on behalf of people living with dementia: how do surrogate decision-makers decide? J Med Ethics. 2017;43(1): 35–40.
[64] Niemeijer AR, Frederiks BJ, Riphagen II, et al. Ethical and practical concerns of surveillance technologies in residential care for people with dementia or intellectual disabilities: an overview of the literature. Int Psychogeriatr. 2010; 22(7):1129–1142.
[65] Epstein EG, Whitehead PB, Prompahakul C, et al. Enhancing understanding of moral distress: the measure of moral distress for health care professionals. AJOB Empirical Bioethics. 2019;10(2):113–124.
[66] Ellis P. Understanding ethics for nursing students (learning matters). 2020.
[67] Milliken A, Grace P. Nurse ethical awareness: understanding the nature of everyday practice. Nurs Ethics. 2017;24(5): 517–524.
[68] Gidlund KL, Sundberg L. Undisclosed practices of digitalization-A critical analysis of representational practice and power. 2019.
[69] Baudin K, Gustafsson C, Frennert S. Views of Swedish elder care personnel on ongoing digital transformation: cross-sectional study. J Med Internet Res. 2020;22(6):e15450.
[70] Oliver M. The social model of disability: thirty years on. Disabil Soc. 2013;28:1024–1026.
[71] Burchardt T. Capabilities and disability: the capabilities framework and the social model of disability. Disabil Soc. 2004;19:735–751.
[72] Anastasiou D, Kauffman JM. The social model of disability: dichotomy between impairment and disability. J Med Philos. 2013;38:441–459.
[73] Astell AJ, Bouranis N, Hoey J, et al. Technology and dementia: the future is now. Dement Geriatr Cogn Disord. 2019; 47(3):131–139.
[74] Berridge C. Breathing room in monitored space: the impact of passive monitoring technology on privacy in independent living. Gerontologist. 2016;56(5):807–816.
[75] Berridge C, Wetle TF. Why older adults and their children disagree about in-home surveillance technology, sensors, and tracking'. Gerontologist. 2020;60(5):926–934.
[76] Neven L. But obviously not for me': robots, laboratories and the defiant identity of elder test users'. Sociol Health Illn. 2010;32:335–347.
[77] Peine A, Rollwagen I, Neven L. The rise of the "innosumer"—Rethinking older technology users. Technol Forecast Soc Change. 2014;82:199–214.
[78] Winner L. Do artifacts have politics? Daedalus. 1980;109: 121–136.
[79] Latour B, Venn C. Morality and technology. Theory Cult Soc. 2002;19:247–260.
[80] Norman D. The design of everyday things: revised and expanded edition (Basic books). 2013.
[81] Oudshoorn N, Pinch T. 'How users matter', The Co. 2003.
[82] Robinson L, Hutchings D, Corner L, et al. Balancing rights and risks: Conflicting perspectives in the management of wandering in dementia. Health Risk Soc. 2007;9:389–406.
[83] Deen MJ. Information and communications technologies for elderly ubiquitous healthcare in a smart home. Pers Ubiquit Comput. 2015;19(3–4):573–599.

[84] Ghasemi F, Rezaee A, Rahmani AM. Structural and behavioral reference model for IoT-based elderly health-care systems in smart home. Int J Commun Syst. 2019;32(12):e4002.
[85] Kamp A, Obstfelder A, Andersson K. 'Welfare technologies in care work'. 2019.
[86] Hofmann B. Too much technology. BMJ. 2015;350:1–7.